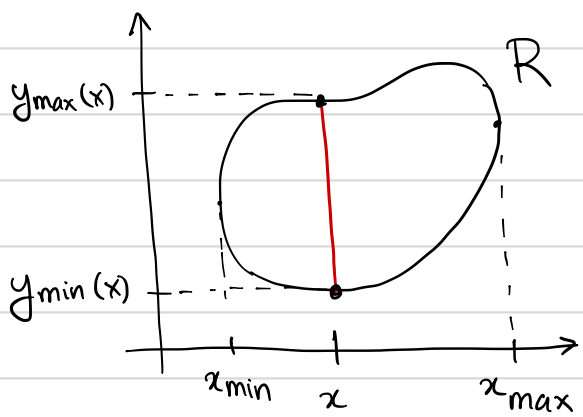
Lecture 2

- Double integrals (15.1-2)
- Applications (15.4)

Course website at

http://math.uw.edu/~lvhb

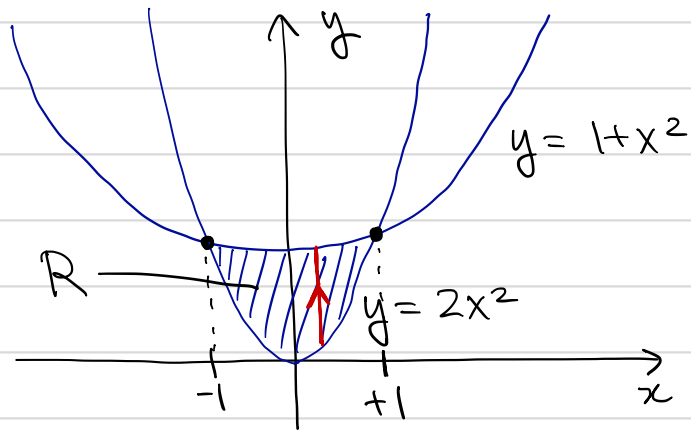
Last time:



Fubini's theorem:

$$\iint_R f(x,y)\,dA = \int_{x_{min}}^{x_{max}} \left[\int_{y_{min}(x)}^{y_{max}(x)} f(x,y)\,dy \right] dx$$

Example: $f(x,y) = x + 2y$



$$\iint_R f(x,y)\,dA = \int_{-1}^{1} \int_{2x^2}^{1+x^2} x + 2y \, dy \, dx$$

$$= \int_{-1}^{1} \Big[xy + y^2 \Big]_{y=2x^2}^{1+x^2} dx$$

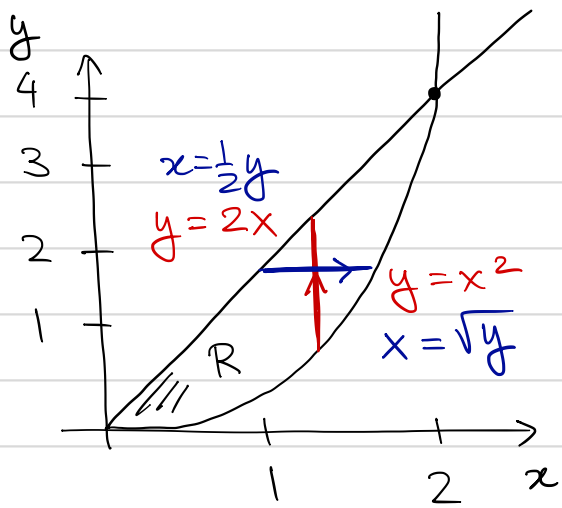
$$= \int_{-1}^{1} \Big[x(1+x^2) + (1+x^2)^2 - 2x^3 - 4x^4 \Big] dx$$

$$= \int_{-1}^{1} \Big[-3x^4 - x^3 + 2x^2 + x + 1 \Big] dx$$

$$= -3\frac{x^5}{5} - \frac{x^4}{4} + 2\frac{x^3}{3} + \frac{x^2}{2} + x \Big|_{x=-1}^{1}$$

$$= \frac{32}{15} \;//$$

Example: $f(x,y) = x^2 + y^2$



$$\iint_R f(x,y)\,dA = \int_0^2 \int_{x^2}^{2x} x^2 + y^2 \, dy\, dx$$

$$= \int_0^4 \int_{y/2}^{\sqrt{y}} x^2 + y^2 \, dx\, dy$$

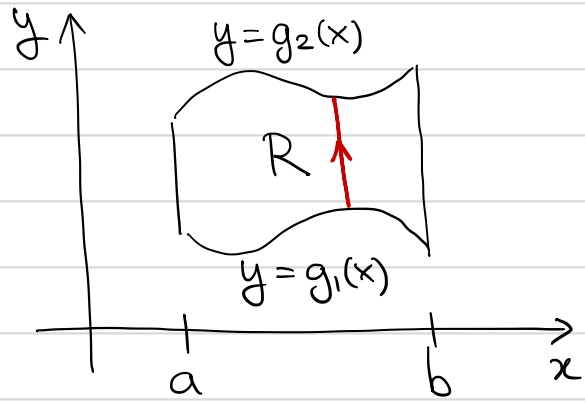
$$= \int_0^4 \left[\frac{x^3}{3} + xy^2 \right]_{x=y/2}^{\sqrt{y}} dy$$

$$= \int_0^4 \left(\frac{y^{3/2}}{3} + y^{5/2} - \frac{y^3}{24} - \frac{y^3}{2} \right) dy$$

$$= \frac{1}{3}\,\frac{2}{5}\, y^{5/2} + \frac{2}{7}\, y^{7/2} - \left(\frac{1}{24} + \frac{1}{2} \right) \frac{y^4}{3} \Big|_0^4$$

$$= \frac{216}{35} \;//$$

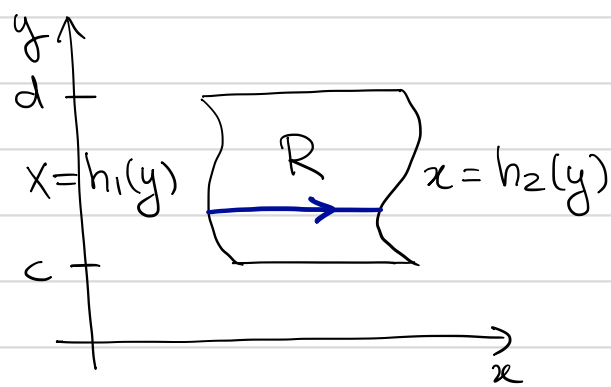
Type I regions (Slides)



$$\iint_R f(x,y)\,dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\,dx$$

$$R = \{ (x,y) \mid a \le x \le b,\ g_1(x) \le y \le g_2(x) \}$$

Type II regions (Slides)

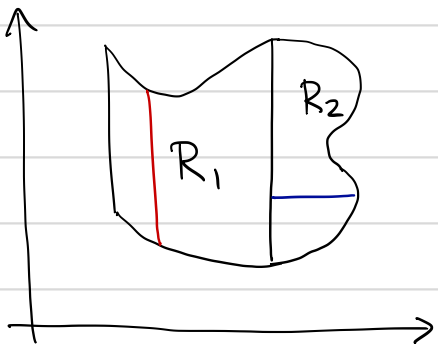


$$\iint_R f(x,y)\,dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)\,dx\,dy$$

$$R = \{ (x,y) \mid c \le y \le d,\ h_1(y) \le x \le h_2(y) \}$$

Some regions are neither type I nor type II:

(Slides)

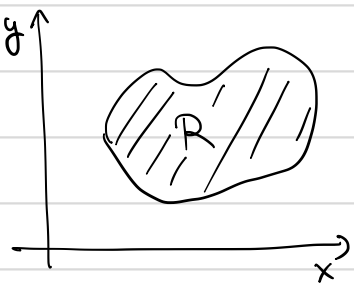


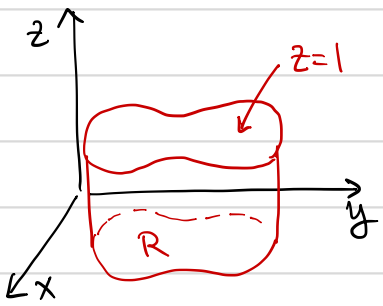
$$\iint_{R_1 \cup R_2} f(x,y)\,dA = \iint_{R_1} f(x,y)\,dA + \iint_{R_2} f(x,y)\,dA.$$

<u>Applications</u>:

1) Find area of region R:

$$\text{Area}(R) = \iint_R 1 \cdot dA$$





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2) Mass of planar object with density ρ = mass per unit area

$$\Delta m = \rho\,\Delta A$$

$$\text{Mass} = \iint_R \rho(x,y)\,dA$$

3) Average value of f in R

$$(\text{Average of } f) = \frac{1}{\text{Area}} \iint_R f\,dA$$

$$\begin{pmatrix}\text{Weighted}\\ \text{average of } f\end{pmatrix} = \frac{1}{\text{Mass}} \iint_R f\rho\,dA$$

<u>Sanity check</u>:

If $m = \min\{f(x,y) \mid (x,y) \in R\}$

$M = \max\{f(x,y) \mid (x,y) \in R\}$

then

$$m \leq \overbrace{\frac{1}{\text{Area}} \iint_R f\,dA}^{\text{average}} \leq M$$

since

$$\underbrace{\iint_R m\,dA}_{m \cdot \text{Area}(R)} \leq \iint_R f\,dA \leq \underbrace{\iint_R M\,dA}_{M \cdot \text{Area}(R)}$$

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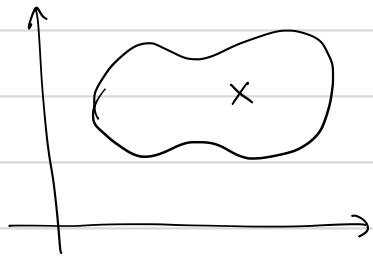
4) Center of mass of a planar object (with density ρ): $(\bar{x}, \bar{y})$



$$\bar{x} = \frac{1}{\text{Mass}} \iint_R x\,\rho(x,y)\,dA$$

$$\bar{y} = \frac{1}{\text{Mass}} \iint_R y\,\rho(x,y)\,dA$$